SF1811 Optimization
Theory results 2026/2027

One of the five questions on the final exam will assess mathematical argumentation and understanding of theory. The question is worth 10 points and consists of two parts.

(a) The first part, worth 5 points, will ask you to formulate and prove, or derive, one of the announced theory results listed below.

(b) The second part, worth 5 points, is not announced in advance. It will ask you to interpret, apply, compare, or delimit the result in part (a).

The two parts are assessed independently. In part (b), you may use the theory result stated in part (a) without proving it.
Full credit for part (a) requires a correct statement of the result, all relevant assumptions, and a logically complete argument without formal or logical deficiencies. A proof may be short, but it must be complete.
The announced theory results are the following.

1. The fundamental theorem of linear programming.

Formulate and prove the fundamental theorem of linear programming.

2. Weak duality for linear programming.

Specify a primal–dual pair of linear programs. State and prove weak duality for this pair. Your statement should include both the inequality between the objective values of arbitrary primal and dual feasible solutions and the resulting inequality between the optimal values.

3. Strong duality for linear programming.

Specify a primal–dual pair of linear programs. State and prove strong duality for this pair. Your statement should include the existence of primal and dual optimal solutions and the relationship between the optimal values. The proof may use the existence of an optimal basic feasible solution and results from the simplex method.

4. Complementarity as an optimality certificate for linear programming. Consider a linear program in standard form and its corresponding dual. Let x be feasible to the primal linear program and let (y, r) be feasible to its dual. Prove that if

$$x_j r_j = 0, \quad j = 1, \ldots, n,$$

then x and (y, r) are optimal solutions to the primal and dual problems, respectively. Your proof may use weak duality.

5. Local and global minimizers in convex optimization.

Assume that $\hat{x}$ is a local minimizer of a convex optimization problem. Prove that $\hat{x}$ is also a global minimizer.

6. First- and second-order necessary optimality conditions.

Assume that $\hat{x}$ is a local minimizer of $f : \mathbb{R}^n \to \mathbb{R}$ and that f is twice continuously differentiable. Prove that

$$\nabla f(\hat{x}) = 0 \qquad \text{and} \qquad F(\hat{x}) \succeq 0,$$

where $F(\hat{x})$ denotes the Hessian of f at $\hat{x}$. Your proof should include the corresponding one-dimensional necessary conditions; these conditions may not be cited without proof.

7. First-order characterization of an unconstrained convex minimizer.

Assume that $f : \mathbb{R}^n \to \mathbb{R}$ is convex and continuously differentiable. Prove that $\hat{x}$ is a global minimizer of f on $\mathbb{R}^n$ if and only if

$$\nabla f(\hat{x}) = 0.$$

You may use the first-order characterization of convexity,

$$f(y) \geq f(x) + \nabla f(x)(y - x), \qquad x, y \in \mathbb{R}^n,$$

without proof.

8. Sufficiency of the KKT conditions for convex problems.

Consider

$$(P) \qquad \begin{array}{lll} \text{minimize} & f(x) & \\ \text{subject to} & h_i(x) = 0, & i \in \mathcal{E}, \\ & g_i(x) \leq 0, & i \in \mathcal{I}, \end{array}$$

where f and all g_i are convex and continuously differentiable and all h_i are affine. The KKT conditions are given on the formula sheet. Prove directly that any point $\hat{x}$, together with multipliers $\hat{u}$ and $\hat{y}$, satisfying the KKT conditions is a global minimizer of (P).

9. Weak duality for Lagrangian relaxation.

The Lagrangian, the dual function, and the Lagrangian dual problem are given on the formula sheet. Formulate and prove weak duality for Lagrangian relaxation.

10. Global optimality conditions.

The global optimality conditions are given on the formula sheet. Formulate and prove the theorem stating that if $(\hat{x}, \hat{u}, \hat{y})$ satisfies these conditions, then $\hat{x}$ is a global minimizer of the original problem.